Description

Definitions:

Application:	executable software, in windows a program ended by *.exe
Algorithm:	sequence of operations.

1 Program structure

Software program is composed by 2 parts:

- Data
- Code, operations on data.

Example: program execute ((**A** + **B**) / 5) * 3. In this case data: {**A, B**}
Data and the code are stored in different memory areas:

- The code manipulates data (sometimes through user intervention), so generally data is read/write.
- The code can't be changed, unless updates are done.

How it works
The program contains instruction in a specific language like C++/C, etc….

1.1 Compiler

So, taking the example above:

- The program will have two variables: **A** and **B**. These variables will be stored in program data memory.
- In order to execute the operation, the code needs access memory where variables **A** and **B** – memory address where they are stored. Something like this:

Memory address	Content	Variable
0x10000010	23	A = 23
0x10000014	52	B = 52

Sequence:

- Read memory 0x10000010 content, which is 23 and store it to A, so A = 23
- Read memory 0x10000014 content, which is 52 and store it to B, so B = 52
- Create temporary variable Tmp = A + B = 23 + 52 = 75
- Calculate the next instructions:
  - Tmp = Tmp / 5 = 75 / 5 = 15
  - Tmp = Tmp * 3 = 15 * 3 = 45

The compiler role is to convert the language code into executing the instructions above, so the CPU can execute the instructions as above.

NOTE: example above expects variables A and B are stored as 4 bytes:
A = [0x10000010…0x10000013] = [0 0 0 23]
B = [0x10000014…0x10000017] = [0 0 0 45]
However, 1 byte could store values: [0, 255]. The A and B could use only 1 memory byte instead of the 4, as presented above. In the later case, the memory would become:

Memory address	Content	Variable
0x10000010	23	A = 23
0x10000011	52	B = 52

How many memory bytes are allocated for variables A and B is determined by the programmer and the programming language. Depending what A and B variables represents in real life, A and B could have different representation.

Variable A 4 bytes memory values: [0, …, 4294967295] and 1 byte memory values: [0, …, 255].

When A represents a environment/house temperature, max 255 Celsius is good enough. A temperature for melting metals could be 1000 Celsius. However, when A represents distances in galaxy, A will need 4 or more bytes of memory to store the value.

The programmer decides A variable type and the compiler will allocate the required memory space to store it.

1.2 Program Data

Native C/C++ language supports the following types of data:

- Numbers data, essentially integers and floating
- Alphanumeric data, string. Sample “This is my house”.

1.2.1 Integer Data

The integer type is the same as defined by math. The differences are related to how they are stored to memory and CPU type:

- Base
- Integer types based on of memory

1.2.1.1 Integer Base

Decimal: 52 = 5 * 10 + 2
Hex 52 = 3 * 16 + 4 = 0x34
Binary: 52 = B 0011 01 00
Numbers are stored in memory in binary format but programming languages use mostly hex format.

1.2.1.2 Integer types

There are the following integer types:

- Char: 1 byte. Valid values: [0,…, 255] or [-128,…, 127]
- Short: 2 bytes. Valid values: [0, …, 65535] or [-32,767, …, 32,768]
- Int: 4 bytes. Valid values: [0, …, 4294967295] or [2,147,483,647, …, 2,147,483,648]
- Long: CPU 32 bytes = 8 bytes,

The memory and CPU does not know how to interpret a number. For example show variable value depending on the type.

Memory address	Value	A type	A value
0x10000010	0x82	Unsigned char	130
0x10000010	0x82	Char	-126

1.2.1.3 TestInt Program

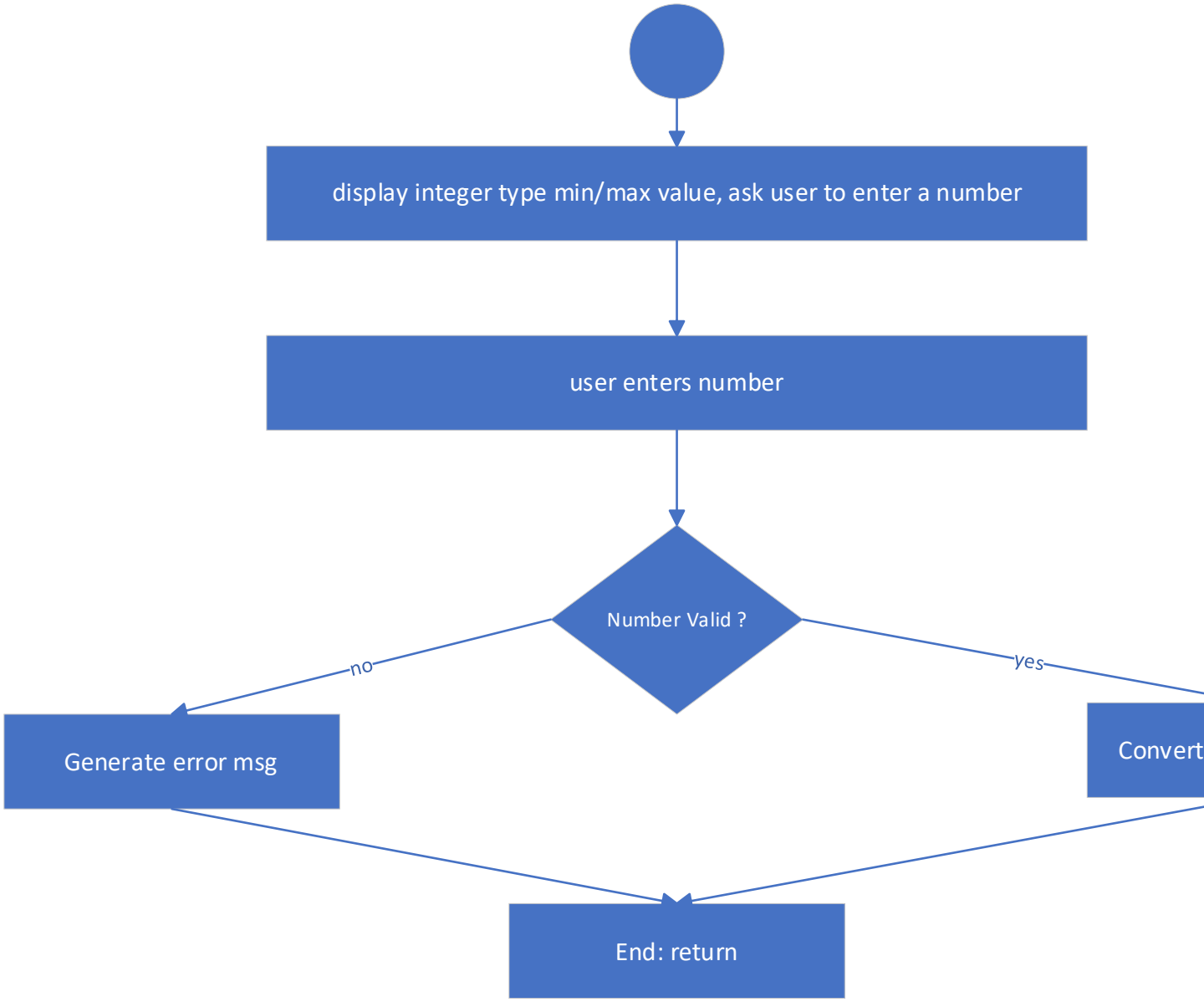
Program TestInt tests integer types and defines one function per type of supported int.

All C/C++ programs are started by function main() executing the sequence of testing various integer types. For each integer type there is a function as follows:

Function	C/C++ type	Definition
processChar()	char	Unsigned, 1 byte
processUChar()	unsigned char	1 byte, [0,…, 255]
processShort()	Short	Unsigned, 2 bytes
…. same rule for other types		

processChar() is analyzed next. The input for each function is the console, defined by C++ language as: **std::in**
The output is the display, defined by C++ language as: **std::out**
The output is the display, defined by C++ language as: **std::err**. In this program is the console.

The input from console is interpreted by computer as a string. So, for example <1234> input is interpreted as string “1234”. No arithmetic operations can be executed on strings. Then the program has to convert the string to integer before executing any specific integer operations.



The integer number Is displayed in 3 formats: decimal, hex and binary. Binary representation is used sometimes in engineering application but it is not used much

1.2.2 Strings

1.2.2.1 General

Programming languages support string data type. A string is a printable data type using human natural alphabet plus some control characters. A string could store a word, one or more phrases, etc…

The string data type is very useful as user interface to the computer. User entered data from the console is interpreted as string – like numbers - and has to be converted to numeric format. The computer output (display) converts numbers to string before being displayed.

Operations on the strings are:

- Initialization, string construction
- String management.
- Copying, appending strings, etc…
- Find and search pattern inside string.

The string internal structure is described below.

The string is a collection of characters. For example, for all programming languages a phrase like “Hello world!” is treated as a string where each character format is ASCII one – see ASCII table at https://www.alpharithms.com/ascii-table-512119. The string is ended by 0x0.

Programming languages uses various variable type string**, C language** definition is presented below as it describes internal string structure:

char str[20] = “Hello”;

String <str> is a table of characters stored in memory as a contiguous area of memory, where each character size is 1 byte. Let say <str> memory address is 0x800000. Then

H	e	l	L	o	**0x0**	A	B	3	0x0	…

Above memory representation, human character and internal value in hex follows:

Memory address	Character	Hex value
0x800000	H	0x48
0x800001	E	0x65
0x800002	l	0x6C
0x800003	l	0x6C
0x800004	o	0x6F
0x800005	**0x0**	**0x0**
…….		

Then, memory 0x800000 contains { 0x48, 0x65, 0x6C, 0x6C, 0x6F**, 0x0**, …….}

1.2.2.2 C language

C language string operations

Check example for oeprations

1.2.2.2.1 Initialization

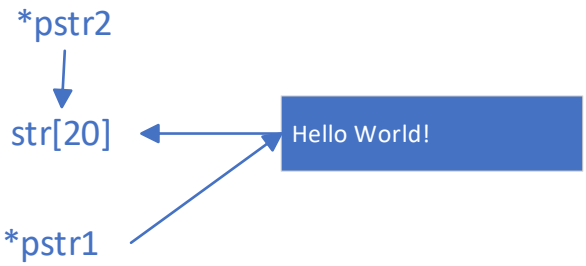
Description:

Three variables are defined:

char str[20] = “Hello World!”; message “Hello World!” is copied to str. str length = 12 , size =20

char *pstr1 = “Hello World”; pstr1 stores the message address.

char *pstr2 = str; pstr2 stores the address of str, which contains message



Executing the following operations:

std::cout << str << “\n” displays “Hello World!” from str.
std::cout << pstr1 << “\n” displays memory “Hello World!”
std::cout << pstr2 << “\n” displays the same as str.

Change str value:

memcpy(str, “Hello Canada!”) ; copy new string “Hello Canada!” to variable <str>. **NOTE: str size is 20 bytes and there is enough space.**

std::cout << str << “\n” displays “Hello Canada!”.
std::cout << pstr1 << “\n” displays memory “Hello World”
std::cout << pstr2 << “\n” displays “Hello Canada!”

NOTE: str variable size is 20 bytes and there is enough space to store the new string. Trying to write as string bigger than its capacity generates “**buffer overflow**” bug which could crash the application and could be used for hacking

1.2.2.2.2 String management

A string has two important parameters:

- String capacity, as defined when string was created. In C language the capacity is fixed and defined at creation time. Example str a[64] defines variable a of type string - a table of 64 characters, last one being 0x0.
- String length. The number of characters currently stored by variable. Function strlen()

Sample:

```
char  a[64];
size_t bytes = strlen(a);  // bytes =0,   no data stored; capacity = 64
char a1[64] = “abc”;
size_t bytes = strlen(a1);          // bytes = 3, capacity = 64
```

The same functions exist in C++, which has other functions too.

1.2.2.2.3 String operations

Copy strings:

strcpy(dst, src); copy source <src> to destination <dst>. **NOTE: <dst> capacity >= <src> size.**

Compare strings:
int strcmp(str1, str2); compare 2 strings

Addition strings
char *strcat(str1, str2); string s2 is appended to string s1. NOTE: <srs1> capacity >= existing <src1> size + <str2> size. Example
    char str1[16] = “Hello “
    char str2[16] = “World!”
   strcat(s1, s2) => s1 = “Hello World!”

1.2.2.2.4 Find and search

Find string:

char *strstr(str1, str2); Find str2 in str1; Example:

   char str1[16] = “Hello World”;

   char str2[16] = “Hello”;

  char *p = strstr(str1, str2); => p = str1[0]